

llmXive Automated Review of Mellum2

Technical Report

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the alignment between claims

and their cited sources.

1. **Benchmark Score Discrepancies and Baseline Verification** In Section 2, Table 1, the paper claims Mellum2 achieves 59.3% on MMLU-Pro, explicitly stating this surpasses Qwen2.5-7B (48.6%). The bibliography cites qwen2.5 (2024) for the Qwen2.5-7B baseline. However, public benchmarks for Qwen2.5-7B on MMLU-Pro often report scores significantly higher than 48.6% (typically in the 50-60% range depending on the specific evaluation protocol). The claim that Mellum2 “surpasses” Qwen2.5-7B relies on this specific 48.6% figure. If the cited source (qwen2.5) reports a higher score, the claim is factually unsupported. The authors must verify that the 48.6% figure corresponds exactly to the version of Qwen2.5-7B cited, or clarify if a different baseline (e.g., a specific fine-tuned version) was used.

2. **Metric Definition in Long-Context Claims** Section 3 states that layer-selective YaRN achieves a “RULER score of 0.64” at 64K context, citing hsieh2024ruler. The RULER benchmark (Hsieh et al., 2024) typically reports results as pass@1 percentages (e.g., 64%) or normalized scores. The notation “0.64” is ambiguous: does it represent 64% or a normalized 0.64? If it is 64%, the text should explicitly state “64%” to avoid confusion with a normalized metric. If it is a normalized score, the citation must support this specific scaling. The current phrasing risks misrepresenting the magnitude of the result relative to the cited work.

3. **Baseline Model Availability and Specificity** Section 5, Table 3, introduces a baseline “Ministral-3-14B” with a LiveCodeBench v6 score of 42.4. The bibliography cites liu2026ministral3 (2026). The “Ministral” family (by Mistral AI) is publicly known for 8B and 3B models. A “14B” variant is not a standard public release in the cited timeframe. The claim that this specific 14B model exists and achieved 42.4% on LiveCodeBench v6 requires verification against the cited source. If the source does not contain a 14B model, the claim is unsupported, and the baseline comparison is invalid.

4. **Data Volume vs. Prompt Count Inconsistency** Section 4 claims the “Thinking” variant was trained on ~167B tokens, while the “Instruct” variant used ~47B tokens. Table 2 lists the prompt counts for the RL data mix as nearly identical (260,500 vs 259,000). While the text mentions “harder long-form math” for the Thinking mix, a 3.5x difference in total training tokens (167B vs 47B) implies a significant difference in average sequence length or data volume not reflected in the prompt counts of Table 2. The text should clarify whether the token count difference arises from longer sequences in the Thinking mix or if the prompt counts in Table 2 are incomplete. Without this clarification, the claim of 167B tokens appears inconsistent with the provided data mix statistics.

Conclusion The paper makes several strong comparative claims that rely on specific baseline scores and model configurations. The discrepancies in MMLU-Pro baselines, the ambiguity of the RULER metric, the existence of the “Ministral-3-14B” model, and the token count vs. prompt count inconsistency require clarification to ensure the claims are accurately supported by the cited evidence.

Action items.

- [science] Verify the MMLU-Pro baseline score of 48.6% for Qwen2.5-7B cited in Table 1.

Public benchmarks often report higher scores for this model. Ensure the cited source (qwen2.5) supports this specific value to validate the claim of surpassing it.

- [writing] Clarify the unit for the RULER score ‘0.64’ in Section 3. The cited paper (hsieh2024ruler) typically reports percentages. Specify if this is 64% or a normalized metric to avoid misinterpretation of the result’s magnitude.
- [science] Confirm the existence and benchmark score of ‘Minstral-3-14B’ in Table 3. The cited source (liu2026minstral3) may not list a 14B variant. Verify if this model and its 42.4% LiveCodeBench score are accurately represented in the reference.
- [writing] Reconcile the 167B token count for the ‘Thinking’ SFT variant with the prompt counts in Table 2. The prompt counts are nearly identical to the ‘Instruct’ variant. Explain if the token difference stems from longer sequences or unlisted data volume.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure is visually clear but the caption is misleading as it claims to show ‘Throughput’ while the axis displays ‘Latency’. Additionally, the plot is cluttered with numerous unlabeled data points that are not defined in the legend or caption.

Figure 2

The figure is a latency contour map that contradicts its caption, which claims to show throughput and concurrent requests. Additionally, the data points plotted on the grid are not defined in the caption or legend, making the visualization ambiguous.

Figure 3

The figure effectively visualizes latency trends but fails to support the caption’s claim of comparing multiple MoE models, as the other configs are not explicitly identified as MoE. Additionally, the label for the candidate configs suffers from poor contrast.

Figure 4

The figure effectively displays the validation loss curves for different optimizers, but the caption is insufficiently descriptive, and the annotation regarding the diverged run is illegible due to poor contrast.

Figure 5

Figure 5 is a clear and well-structured line plot showing LM loss over training steps. The axes are labeled with units, the three training phases are visually distinguished with background shading and text, and the vertical dashed lines align with phase transitions. The caption accurately describes the figure content.

Figure 6

The figure is clear and the data presentation is effective, but the caption contains raw LaTeX syntax errors in the method names and a broken cross-reference regarding score caveats.

Figure 7

The figure clearly displays the relationship between RULER scores and training tokens for three model sizes, but the caption contains an incomplete cross-reference and the x-axis unit is slightly ambiguous.

Figure 8

The figure clearly displays the load-balancing loss trend over training tokens, but the annotation regarding the ‘RULER plateau’ is undefined, and the specific loss metric is not detailed in the caption.

Figure 9

Figure 9 clearly displays training and validation accuracy curves with appropriate axis labels, units, and a legend. The caption accurately describes the visualization, including the smoothed train curve, raw per-step values, and validation sampling frequency.

Figure 10

The figure effectively presents the benchmark data with clear axes and labels, but the caption contains a grammatical error omitting the subject, and the visual representation of the latency comparison lacks error bars to substantiate the claim of matching performance. Action items.

- [science] Figure 1: The caption ‘Throughput mode’ contradicts the y-axis label ‘Latency (s)’; the figure displays latency, not throughput.
- [writing] Figure 1: The legend defines ‘Mellum 4B’ and ‘Qwen2.5-7B’ but the plot contains

dozens of unlabeled data points (e.g., ‘T:11.6B’, ‘A:2.5B’) that are not defined in the caption or legend.

- [writing] Figure 1: The x-axis label ‘Depth (Layers)’ is ambiguous as the tick marks (24, 26, 28...) do not clearly correspond to the specific model configurations labeled on the plot.
- [science] Figure 2: The caption describes ‘Throughput mode (concurrent requests)’, but the plot axes are ‘Width (Hidden Dimension)’ and ‘Depth (Layers)’, and the colorbar is ‘Latency (s)’. The figure displays a latency contour map, not throughput or concurrent request data, contradicting the caption.
- [science] Figure 2: The legend identifies ‘Mellum 4B’ and ‘Qwen2.5-7B’, but the plot contains a dense grid of data points (grey circles) with specific values (e.g., ‘7.05B’, ‘8.62B’) that are not explained in the caption or legend. It is unclear if these points represent the models or a parameter sweep.
- [science] Figure 3: The caption claims to compare ‘MoE models’ (plural) with Sliding Window Attention, but the plot only explicitly labels one MoE model (‘Mellum 2’) and a group of ‘6 other candidate configs’ without specifying if they are MoE or their window sizes. This makes the claim of comparing ‘MoE models’ (plural) unsupported by the visual data.
- [writing] Figure 3: The text ‘6 other candidate configs’ is rendered in a very light gray font that is difficult to read against the white background, reducing legibility.
- [science] Figure 4: The caption ‘Qwen2.5-7B (dense)’ is too generic and fails to describe the specific comparison shown (Muon vs. AdamW vs. Muon Moonlight) or the metric (Validation loss), making the figure’s purpose unclear without external context.
- [writing] Figure 4: The top annotation ‘Muon (Megatron defaults) — diverged -> 2.47 at 21B’ is rendered in a very small, light orange font that is difficult to read against the white background.
- [writing] Figure 4: The legend labels ‘AdamW’ and ‘Muon (Moonlight)’ are placed directly on the plot area without a distinct legend box or leader lines, which can be confusing if the lines were to cross or overlap.
- [writing] Figure 6: The caption contains raw LaTeX formatting artifacts (‘uniform
—bump'and'unchanged—
' instead of the readable method names (‘Uniform RoPE base’ and ‘Unchanged RoPE base’) shown in the plot labels.
- [writing] Figure 6: The caption references ‘See for caveats on the absolute scores’ but lacks the specific figure or section number to which the reader should refer.
- [writing] Figure 7: The caption references ‘See for a comment on absolute RULER scores’ but does not specify which section, figure, or table to consult.

- [science] Figure 7: The x-axis label ‘Long-context training tokens (B)’ is ambiguous; it is unclear if ‘B’ refers to billions of tokens or a specific batch size parameter.
- [writing] Figure 8: The annotation ‘RULER plateau (≈ 30 B)’ is present but lacks a definition in the caption or legend explaining what the ‘RULER plateau’ signifies in the context of load-balancing loss.
- [science] Figure 8: The y-axis label ‘Load-balancing loss’ is generic; the caption does not specify the exact loss function or metric used (e.g., auxiliary loss coefficient, specific MoE load-balancing formula), making it difficult to interpret the absolute values.
- [writing] Figure 10: The caption contains a sentence fragment (‘matches the sync latency...’) that lacks a subject, failing to explicitly state that ‘Mellum 2’ is the model being described.
- [science] Figure 10: The ‘Sync (single request)’ chart displays raw values (193 vs 192) but does not visually represent the ‘matches’ claim in the caption, as the bars are distinct in length without error bars to indicate statistical equivalence.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific acronyms and proprietary or niche terminology that are not defined upon first use, creating a barrier for non-specialist readers.

In the Abstract and Section 2 (Model Architecture), terms like Muon, YaRN, MTP, GQA, and SWA are introduced as acronyms or proper nouns without their full expansions or explanations. For instance, “Muon optimizer” and “YaRN” are critical to the methodology but remain undefined. Similarly, Section 3 (Long Context Extension) uses “FIM-formatted” without defining FIM (Fill-in-the-Middle).

Section 4 (Post-Training) introduces RLVR (Reinforcement Learning with Verifiable Rewards) and SFT (Supervised Fine-Tuning) without explicit definitions in the abstract or early text. Most critically, Section 4.2.3 introduces “IcePop truncation” as a specific guard mechanism. “IcePop” appears to be an internal project name or a very specific reference not explained in the text, leaving the reader unable to understand the mechanism’s nature or origin.

Additionally, the abbreviation “pp” (percentage points) is used in Section 3.1 and 3.3 without definition. While standard in some circles, it should be spelled out or defined for a general technical report.

To improve accessibility, the authors should ensure every acronym is defined at its first occurrence (e.g., “Grouped-Query Attention (GQA)”) and that proprietary or internal terms like “IcePop” are either explained or replaced with descriptive text.

Action items.

- [writing] The manuscript relies heavily on domain-specific acronyms and proprietary or niche terminology that are not defined upon first use, creating a barrier for non-specialist readers. In the Abstract and Section 2 (Model Architecture), terms like Muon, YaRN, MTP, GQA, and SWA are introduced as acronyms or proper nouns without their full expansions or explanations. For instance, “Muon optimizer” and “YaRN” are critical to the methodology but remain undefined. Similarly, Section 3 (Long Context Extensi

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper generally maintains a strong logical flow between the architectural choices (MoE, GQA, MTP) and the stated goals of efficiency and code specialization. The causal chain from the three-phase pre-training curriculum to the specific benchmark results is well-articulated, particularly in the ablation studies for long-context extension (Section 3).

However, there are minor inconsistencies in the presentation of results versus claims. In the Abstract and Introduction, the authors state the model “matches” the throughput of Qwen2.5-7B. However, Section 6 (Efficiency and Deployment) explicitly reports a 21% *higher* throughput (20.2 req/s vs 16.7 req/s). While “matching” is technically true if one considers the baseline, the phrasing obscures a positive result and creates a slight logical dissonance with the data presented later.

Additionally, the description of the RL algorithm in Section 4.3 presents a potential logical gap. The text explicitly states “No KL term to SFT reference,” which is a significant deviation from standard PPO/GRPO formulations. However, the loss function includes a masking term $M(\rho)$ based on the ratio of training to inference probabilities. While this is a valid technique (IcePop truncation), the paper does not explicitly explain how this mechanism replaces or interacts with the missing KL divergence term to prevent policy collapse. The reader is left to infer the stability mechanism, which weakens the logical completeness of the method description.

Finally, the description of the “Thinking” variant’s RL training in Section 5.2 mentions a “cold restart,” but the evaluation tables imply a sequential progression. Clarifying whether the RL stage is a fresh initialization or a continuation of the SFT weights would strengthen the logical consistency of the training narrative.

Action items.

- [writing] The claim that the model ‘matches Qwen2.5-7B throughput’ (Abstract, Intro) is contradicted by the specific metric in Section 6, which states Mellum2 achieves ‘21% higher throughput’ (20.2 req/s vs 16.7 req/s). The conclusion should reflect the actual

performance gain rather than a match.

- [writing] In Section 4.3 (RL algorithm), the text states ‘No KL term to SFT reference,’ yet the loss formula includes a masking term $M(\rho)$ derived from the ratio of training to inference probabilities. The logical link between the absence of a KL penalty and the presence of this specific truncation mechanism needs explicit clarification to avoid confusion about the objective function.
- [writing] Section 5.2 claims the ‘Thinking’ variant uses a ‘cold restart from SFT-Thinking’ for RL, but the evaluation tables (Table 7) show a progression from SFT to RL. The logical flow of the training pipeline (Cold restart vs. continued fine-tuning) should be explicitly defined to ensure the reported gains are correctly attributed to the RL stage.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several strong claims regarding efficiency and performance that slightly overreach the provided evidence or lack necessary context for verification.

First, the efficiency claims in the Introduction and Section 6 are the most vulnerable to over-interpretation. The authors state the model “matches Qwen2.5-7B throughput on H100” and achieves “21% higher throughput” (20.2 req/s vs 16.7 req/s). However, the text does not specify the batch size, sequence length, or specific vLLM configuration used to generate these numbers. Throughput is a function of these variables; without them, the claim that the architecture *inherently* matches the baseline is not fully substantiated. The comparison could be an artifact of a specific, non-standard inference setup rather than a general architectural advantage.

Second, the Abstract and Introduction claim the model is “competitive with 4B–14B open baselines at 2.5B dense compute.” While the throughput data supports the “2.5B compute” aspect, the performance data in Table 1 (Pre-training evaluation) shows the model lagging behind Qwen2.5-7B on key benchmarks like HumanEval (41.5% vs 55.5%) and MMLU (70.9% vs 71.8%). The term “competitive” is used loosely here; the model is faster but less accurate on these specific tasks. The paper should qualify this claim to reflect the specific trade-off (speed vs. accuracy) rather than implying a general equivalence in capability.

Third, the causal attribution of training stability to the “Muon – FP8 recipe” (Abstract, Section 1) is an over-extrapolation. While the paper details the use of these techniques, it does not provide an ablation study isolating their effect from the MoE architecture or the data curriculum. Attributing the stability of 10.6T token training solely to the optimizer and precision format ignores other potential factors (e.g., data quality, MoE routing stability) that are not ruled out.

Finally, regarding the long-context extension, the authors admit that “Absolute scores are conservative due to a prompt-formatting issue” (Section 3). While they correctly highlight the *relative* improvement of layer-selective YaRN, the phrasing “outperforms uniform RoPE” could be misread as a claim of absolute state-of-the-art performance. The text should be tightened to emphasize that the *ranking* is robust despite the absolute score suppression, preventing readers from over-estimating the model’s raw long-context capability.

Action items.

- [science] The claim that Mellum2 ‘matches Qwen2.5-7B throughput on H100’ (Sec 1) and ‘matches sync latency’ (Sec 6) is over-claimed without explicit hardware configuration details (e.g., batch size, sequence length, precision mode) in the text. The throughput figure (20.2 req/s) is highly sensitive to these parameters; without them, the comparison is not reproducible or verifiable.
- [writing] The statement that the model is ‘competitive with 4B–14B open baselines at 2.5B dense compute’ (Abstract) over-extrapolates the results. While it matches Qwen2.5-7B in throughput, the pre-training evaluation (Table 1) shows it underperforming Qwen2.5-7B on HumanEval (41.5% vs 55.5%) and MMLU (70.9% vs 71.8%). The ‘competitive’ claim requires qualification regarding the specific trade-off between speed and accuracy.
- [science] The assertion that ‘Muon – FP8 recipe’ enabled ‘Stable training at 10T tokens’ (Sec 1) is a causal claim not fully supported by the provided ablation data. The paper mentions Muon and FP8 but does not present a controlled ablation isolating these factors from the MoE architecture or data curriculum to prove they were the sole or primary drivers of stability at that scale.
- [writing] The claim that ‘Layer-selective YaRN... outperforms uniform RoPE base’ (Sec 3) is supported by RULER scores, but the text admits ‘Absolute scores are conservative due to a prompt-formatting issue’ (Sec 3). The paper should explicitly state that the *relative* improvement is the valid finding, rather than implying the absolute performance is fully realized, to avoid over-claiming the model’s actual long-context capability.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript presents a technical report on the Mellum2 model, an open-weight MoE language model. From a safety and ethics perspective, the paper is generally transparent about its training pipeline and evaluation metrics, including specific safety benchmarks (HarmBench, XSTest). However, there are critical omissions regarding the interpretation of safety results and the potential risks associated with the model’s specific capabilities.

First, the “Post-Training Evaluation” section (Section 5) explicitly notes a safety regres-

sion: the HarmBench refusal rate drops significantly (score increases from 8.4% to 23.1%) after Reinforcement Learning (RL) is applied to the Instruct variant. While the authors report this number, they do not provide an analysis of *why* this degradation occurred or how it impacts the model’s deployment safety. Given that the model is released under Apache 2.0, users need a clear warning that the RL-tuned variant may be more susceptible to jailbreaking or generating harmful content compared to the SFT-only base. This requires a dedicated discussion in the safety section or a prominent disclaimer in the model card.

Second, the paper introduces a “Thinking” variant that emits reasoning traces. The RLVR process involves optimizing for these traces. There is a potential dual-use risk where the model might learn to generate harmful reasoning steps (e.g., planning a cyberattack or detailing a chemical synthesis) that are technically “correct” according to the reward function but ethically problematic. The paper does not discuss whether the safety filters or the reward model specifically penalize harmful content *within* the reasoning trace, or if the “Thinking” variant is more prone to leaking such content than standard models.

Third, the RL infrastructure relies on a “Verification stack” that includes code execution (Section 4.3.2). While the paper mentions a “decoupled” stack, it lacks details on the security of the execution environment. If the model is rewarded for generating code that passes tests, there is a risk it could generate code that performs malicious actions (e.g., network calls, file system manipulation) if the test harness is not strictly sandboxed. The authors should confirm that the code execution environment is air-gapped or strictly sandboxed to prevent the training loop itself from being a vector for harm.

Finally, the data composition includes “Identity examples (oversampled 3x to prevent misidentification as Google AI)” (Section 4.1.1). While this addresses a specific identity issue, the paper does not discuss the broader data privacy implications of the pre-training corpus (10.6T tokens) or the SFT data, particularly regarding the potential inclusion of personally identifiable information (PII) or copyrighted code. A brief statement on data filtering for PII and copyright compliance would strengthen the ethical standing of the report.

Action items.

- [writing] The paper reports a significant safety regression in the RL-trained ‘Instruct’ variant (HarmBench score increases from 8.4% to 23.1%, Section Post-Training Evaluation). The authors must explicitly discuss the mechanism of this degradation and provide a mitigation strategy or a clear warning for users deploying the RL variant in sensitive environments.
- [writing] The ‘Thinking’ variant is trained on a mix including ‘Reasoning’ and ‘Math’ with RLVR. The paper does not address the risk of these models generating harmful content within their ‘reasoning traces’ (chain-of-thought) which might be visible to users or logged, potentially bypassing standard safety filters that only inspect final outputs.
- [writing] The RL data mix includes ‘Agentic tool use’ and ‘Code’ with programmatically verifiable rewards. The authors must clarify if the code execution sandbox used for reward verification is fully isolated to prevent the model from executing malicious

payloads (e.g., data exfiltration, system compromise) during the training loop itself.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a comprehensive technical report on the Mellum2 model, detailing a 12B-parameter MoE architecture, extensive pre-training, and post-training via SFT and RLVR. The scale of the training data (~10.6T tokens) and the complexity of the pipeline (Muon optimizer, FP8, layer-selective YaRN) are well-documented. However, the scientific evidence supporting the specific efficacy of the proposed techniques and the robustness of the reported benchmarks requires strengthening.

First, the evaluation methodology lacks statistical rigor. Tables 1, 3, and 4 present benchmark results (e.g., HumanEval, MMLU-Pro, LiveCodeBench) as single scalar values. In the absence of reported standard deviations, confidence intervals, or the number of evaluation seeds (e.g., pass@1 over 100 runs), it is impossible to determine if the observed differences between Mellum2 and baselines (e.g., the 10.7 point gap in MMLU-Pro) are statistically significant or within the noise floor of the evaluation. For a technical report making competitive claims, reporting variance is essential.

Second, the causal attribution of performance gains in the Reinforcement Learning (RL) section is weak. The authors introduce specific mechanisms like “IcePop truncation” to handle MoE router non-determinism and “concision penalties” to reduce verbosity. While the final RL scores show improvement over SFT, the paper does not provide an ablation study isolating these components. Without a controlled experiment comparing the full RL setup against a baseline GRPO implementation without these specific modifications, the claim that these specific techniques drove the observed improvements remains an assertion rather than a demonstrated fact.

Finally, the long-context evaluation contains a caveat that undermines the primary metric. The authors report a RULER score of 0.64 at 64K context but acknowledge in Appendix A.1 that a “prompt-formatting issue” caused the model to fail QA subsets, making the scores “conservative.” If the reported 0.64 is indeed an underestimate, the claim that the model is competitive at 128K context is not fully supported by the provided numbers. The authors should either provide corrected scores or explicitly state that the current figures are lower bounds, adjusting their comparative claims accordingly.

Action items.

- [science] Table 1 (Pre-training) and Tables 3-4 (Post-training) report single-point benchmark scores without confidence intervals, standard deviations, or seed counts. For claims of statistical significance (e.g., MMLU-Pro 59.3% vs 48.6%), the authors must report the number of evaluation seeds and variance to rule out random fluctuation.

- [science] The RLVR section (Sec 4.2) claims performance gains from ‘IcePop truncation’ and ‘concision penalties’ but provides no ablation study isolating these specific components from the base GRPO algorithm. Without a controlled comparison (e.g., GRPO vs. GRPO–IcePop), the attribution of gains to these specific mechanisms is unsupported.
- [science] The long-context extension (Sec 3) cites a RULER score of 0.64 at 64K but notes a ‘prompt-formatting issue’ in the appendix. The authors must clarify if the reported scores are corrected for this issue or if the 0.64 figure is an underestimate. If the latter, the claim of ‘competitive’ long-context performance requires re-evaluation with corrected metrics.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents a comprehensive technical report on the Mellum2 model, but the statistical rigor of the evaluation section requires significant strengthening to support the comparative claims.

Lack of Variance Reporting: The primary statistical deficiency is the reporting of benchmark results as single-point percentages without any measure of uncertainty (standard deviation, standard error, or confidence intervals). This is evident in Table 1 (Pre-training evaluation), Table 4 (Post-training Instruct), and Table 5 (Post-training Thinking). For instance, the claim that Mellum2 achieves 59.3% on MMLU-Pro compared to Qwen3.5-4B’s 52.4% is presented as a definitive fact. However, without knowing the variance across multiple evaluation runs or seeds, it is impossible to determine if this 6.9 percentage point difference is statistically significant or within the noise margin of the benchmark. This is particularly critical for benchmarks like HumanEval (41.5% vs 45.1%) where the differences are smaller and the pass@k metric inherently has high variance.

RL Evaluation Methodology: In Section 5 (Post-Training), the Reinforcement Learning (RL) results are presented without methodological details regarding statistical stability. The tables show SFT vs. RL improvements (e.g., LiveCodeBench v6 jumping from 30.9% to 37.2% in Instruct mode). The text does not specify:

1. The number of independent training seeds used.
2. Whether the reported scores are averages over multiple inference runs with different sampling temperatures or seeds.
3. The confidence intervals for these improvements.

Given the stochastic nature of RL training and inference, a single run result is insufficient to claim a robust improvement. The “Thinking” variant’s performance drop on Live-

CodeBench (75.1% to 69.9%) after RL also lacks context on whether this is a statistically significant regression or a fluctuation.

Throughput Claims: In Section 6 (Efficiency and Deployment), the authors claim Mellum2 achieves “21% higher throughput” (20.2 req/s vs 16.7 req/s) compared to Qwen2.5-7B. This is a precise quantitative claim that requires statistical backing. The manuscript does not state the sample size (number of requests processed), the duration of the measurement, or the variance in throughput measurements. Without this, the claim could be influenced by transient system load or measurement noise.

Recommendations:

1. Re-run evaluations with multiple seeds (at least 3-5) for all key benchmarks and report mean ± standard deviation.
2. Add confidence intervals (e.g., 95% CI) to the benchmark tables or explicitly state the statistical significance (p-values) of the differences between Mellum2 and baselines.
3. Clarify the inference protocol (temperature, top-p, number of samples) used for generating the reported scores.
4. Provide statistical details (N, variance) for the throughput comparison in Section 6.

Until these statistical details are provided, the comparative claims regarding performance superiority or parity remain scientifically unsupported.

Action items.

- [science] Report uncertainty metrics (standard deviation or confidence intervals) for all benchmark scores in Tables 1, 4, and 5. Single-point estimates without variance measures prevent assessment of statistical significance, especially for small margins (e.g., MMLU-Pro 59.3% vs 58.6%).
- [science] Clarify the statistical methodology for RLVR evaluation. Specify the number of independent seeds, sampling temperature, and whether results are averaged over multiple runs or single deterministic passes. The lack of variance reporting in RL tables (e.g., LiveCodeBench 75.1% vs 69.9%) obscures result stability.
- [science] Define the statistical basis for the ‘21% higher throughput’ claim in Section 6. Provide the sample size (number of requests), confidence interval, or p-value supporting this difference against the Qwen2.5-7B baseline to rule out random fluctuation.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a technically dense and ambitious report on the Mellum2 model.

However, the writing quality is currently compromised by significant structural fragmentation and inconsistent narrative flow, likely resulting from the assembly of the source text.

The most critical issue is the apparent duplication and splitting of major sections. Specifically, the content regarding “Long Context Extension” and “Post-Training” appears in disjointed blocks (e000 vs. e001/e002). This results in repeated figure captions, table definitions, and explanatory text. For instance, the description of the YaRN extension and the associated RULER scores is presented twice with slight variations. This fragmentation disrupts the reader’s ability to follow the logical progression of the research. The authors must consolidate these sections into a single, coherent narrative, ensuring that each figure and table is defined and discussed exactly once.

Additionally, the transition between the architectural design decisions and the benchmark results in Section 3.1 is abrupt. The text moves from a discussion of dense vs. sparse configurations directly into a large table without a clear introductory sentence explaining the context of the comparison. A brief lead-in sentence would significantly improve the cohesion of this section.

In Section 6.3.1, the introduction of the “IcePop truncation” mechanism is somewhat opaque. The text introduces the variable ρ_t and the parameters α and β within the context of the loss formula without explicitly defining them in the preceding prose. While the formula is provided, the text should clearly state what these variables represent (e.g., “where ρ_t is the train-vs-inference ratio...”) to ensure the reader can understand the mechanism without reverse-engineering the math.

Finally, several figure captions are incomplete, ending with ellipses or fragmented phrases (e.g., “See \cref{app:ruler-formatting} for caveats on the absolute scores...”). These should be rewritten as complete, self-contained sentences that fully describe the figure’s content and any necessary caveats. Addressing these structural and stylistic issues will greatly enhance the readability and professional quality of the report.

Action items.

- [writing] The manuscript contains significant structural fragmentation. Section 5 (Long Context) and Section 6 (Post-Training) appear to be duplicated or split across disjointed chunks (e000 vs e001/e002), resulting in repeated figure captions, table definitions, and narrative text. The authors must consolidate these sections into a single, continuous flow to ensure readability.
- [writing] In Section 3.1 (Architecture Design Decisions), the text abruptly transitions from discussing dense vs. sparse configurations to a benchmark table without a clear introductory sentence explaining the table’s purpose or the specific metrics being compared. A brief lead-in sentence is required to improve cohesion.
- [writing] The ‘IcePop truncation’ mechanism is introduced in Section 6.3.1 (RL algorithm) with a reference to an undefined variable ho_t and parameters $lpha, eta$ without prior definition in the text. The manuscript should explicitly define these variables and their physical meaning before presenting the loss formula.

- [writing] Several figure captions (e.g., Fig 2, Fig 3, Fig 4) end with ellipses (...) or incomplete sentences (e.g., '... See \cref{app:ruler-formatting} for caveats...'). These captions must be rewritten as complete, self-contained sentences to meet standard technical writing conventions.